We refer to $\mathcal{M}_f$ as the *adjoint* operator[3] of $\mathcal{L}_f$ and we write $\mathcal{M}_f = \mathcal{L}_f^\dagger$, where

$$\langle u, \mathcal{L}_f v\rangle = \langle \mathcal{L}_f^\dagger u, v\rangle$$

and

$$\mathcal{L}_f^\dagger u = -\nabla \cdot (u\boldsymbol{f}).$$

The density ρ evolves as the solution of the partial differential equation

$$\frac{\partial \rho}{\partial t} = \mathcal{L}_f^\dagger \rho,$$

where we interpret ρ as a function of both $\boldsymbol{z}$ and t and view $\mathcal{L}_f^\dagger$ as only operating on the spatial components. This means, formally, $\rho(\boldsymbol{z}, t) = e^{t\mathcal{L}_f^\dagger}\rho(\boldsymbol{z}, 0)$. We can think of $\exp(t\mathcal{L}_f^\dagger)$ as defining the *propagator* of measure.

5.2.3 The Liouville Equation for Hamiltonian Systems

Until now we have considered a general system $\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z})$. What happens when $\boldsymbol{f}$ is derived from a Hamiltonian?

From the above we know that for a smooth scalar function $u = u(\boldsymbol{z})$,

$$\mathcal{L}_f^\dagger u = -\nabla \cdot (u\boldsymbol{f}) = -\boldsymbol{f} \cdot \nabla u - u\nabla \cdot \boldsymbol{f}.$$

The term $\nabla \cdot \boldsymbol{f}$ vanishes for $\boldsymbol{f} = \boldsymbol{J}\nabla H$. Thus we have

$$\mathcal{L}_f^\dagger u = -\boldsymbol{f} \cdot \nabla u = -\mathcal{L}_f u.$$

If this holds for all smooth u, we write

$$\mathcal{L}_f^\dagger = -\mathcal{L}_f.$$

[3]The adjoint operator (acting on functions) is not to be confused with the adjoint of a map (which acts on points in phase space) encountered in Chap. 2.